

Answer Key — Biology Grade 10

Sub-Strand 1.2: Chemicals of Life • Lesson 2: Elements and Compounds of Life: What Is Wanjiru's Lunch Made Of?

Teacher reference only. Do not project or hand out.

Quick marking grid

Q1	Q2	Q3	Q4	Q5	Q6	Q7
B	B	B	D	C	D	A

USE AFTER: OBSERVE PHASE — Station A: Nutrient data

Q1. Wanjiru's meal has ugali (about 70 g carbohydrate), beans (about 20 g protein and 15 g carbohydrate), sukuma wiki (about 3 g protein plus fibre) and cooking oil (about 10 g fat). What is the total mass of carbohydrate in the meal?

- A. 70 g
- B. 85 g**
- C. 105 g
- D. 95 g

Correct answer: B. Carbohydrate comes from ugali and beans only: $70\text{ g} + 15\text{ g} = 85\text{ g}$. The other values leave out the beans or wrongly add protein or fat.

USE AFTER: OBSERVE PHASE — Station C: Burn test demonstration

Q2. Mr Otieno heats a small piece of dry maize flour in a flame and it turns black. Table salt heated in the same flame does not turn black. What does the black residue show?

- A. The maize flour contains salt, which turns black when heated
- B. The maize flour contains carbon, which is present in organic compounds**
- C. The maize flour contains iron, which turns black when heated
- D. The maize flour contains water, which turns black when heated

Correct answer: B. Blackening (charring) is the carbon left behind when an organic food burns. Salt is inorganic and has no carbon, so it does not blacken.

USE AFTER: OBSERVE PHASE — Station B: Element cards

Q3. Starch (a carbohydrate) is made of C, H and O. Which element is found in proteins, such as those in beans and milk, but is NOT found in carbohydrates like starch?

- A. Oxygen (O)
- B. Nitrogen (N)**
- C. Hydrogen (H)
- D. Carbon (C)

Correct answer: B. Proteins contain nitrogen (and sometimes sulphur) in addition to C, H and O. Carbohydrates contain only C, H and O.

USE AFTER: EXPLAIN PHASE — Classification Chart: Organic and Inorganic columns

Q4. Which list contains ONLY organic compounds found in Wanjiru's meal?

- A. Protein, calcium salts, fat, starch
- B. Starch, water, fat, vitamin C
- C. Water, iron salts, calcium salts, protein
- D. Starch, protein, fat, vitamin C**

Correct answer: D. Carbohydrates (starch), proteins, lipids (fat) and vitamins are organic. Water and mineral salts such as calcium and iron salts are inorganic, so they spoil the other lists.

USE AFTER: EXPLAIN PHASE — Classification Chart: why water goes in the inorganic column

Q5. Water is found in every living thing and in all foods, yet it is placed in the inorganic column. Why?

- A. Water is a liquid and organic compounds are always solids
- B. Water is not a chemical, so it cannot be organic
- C. Water is made of only hydrogen and oxygen and has no carbon bonded to hydrogen**
- D. Water is not found inside cells

Correct answer: C. Organic compounds contain carbon bonded to hydrogen. Water (H and O only) has no carbon, so it is inorganic even though living things need it.

USE AFTER: EXPLAIN PHASE — Claim-Evidence-Reasoning statement

Q6. Wanjiru's blood glucose rose after she ate a meal containing about 70 g carbohydrate in ugali, about 20 g protein in beans and about 10 g fat from cooking oil. Which claim and evidence is best supported by this data?

- A. Water is most likely the cause, because it is found in all foods
- B. Protein is most likely the cause, because beans contain 20 g of it
- C. Fat is most likely the cause, because cooking oil is added to the meal
- D. Carbohydrate is most likely the cause, because the meal contains about 70 g of it in ugali, more than any other nutrient**

Correct answer: D. The largest amount of any nutrient is carbohydrate (about 70 g in ugali), so it is the most likely link to the glucose rise. The other claims are not supported by the quantities given.

USE AFTER: MODEL BUILDING PHASE — Class model flow diagram: filling in the first box, Food chemicals

Q7. In the class model, the first box 'Food chemicals' is filled in, but the box 'Glucose in blood' is left as a question mark. Why?

- A. We know the kinds of chemicals in the lunch, but we do not yet know which one becomes glucose or how**
- B. Carbohydrates have no link to the glucose rise
- C. Glucose is never found in blood
- D. The model is finished and no more boxes are needed

Correct answer: A. Models grow only with evidence. This lesson gives evidence about the groups of chemicals in food, not how they change into glucose, so that box stays open for later lessons.